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SpiceSync

“Fast Food Management System”

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1. Module M1: User Account Management

Covers all user-facing account operations: registration, login, OTP verification, password management, social OAuth, profile editing, language preferences, session management, and account deletion.

1.1 M1-F1: User Registration (Email & Password)

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Email	user@gmail.com, ali.khan@outlook.com	user@, @gmail, plain, null, spaces	64-char local, 255-char full	RFC 5321 compliant
Password	Spice@2025, P@ss#99	12345, pass, abc, empty	8 chars (min), 128 chars (max)	Upper+lower+digit+symbol required
Phone	+92-300-1234567, +92-321-9876543	03001234, +1-ABC, null, 7 digits	+92XXXXXXXXXX (13 chars total)	Pakistan mobile only
Username	ali_99, fatima.noor	ab (too short), ali!!, null, spaces	3 chars (min), 30 chars (max)	Alphanumeric + _ + . only

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M1F1-01	Functional	Successful registration with all valid fields	Email:user@gmail.com PW:Spice@2025 Phone:+92-300-1234567	1.Open app 2.Tap Sign Up 3.Fill valid fields 4.Tap Register	Account created; OTP sent to email; Welcome screen shown	—
TC-M1F1-02	Functional	Registration with minimum required fields	Email + Password only (optional fields skipped)	1.Enter email & password only 2.Tap Register	Registration successful with minimal data; OTP email sent	—
TC-M1F1-03	Negative	Register with already-used email	Email: existing@gmail.com	1.Enter email already in system 2.Fill other fields 3.Tap Register	Error: 'This email is already registered. Please login.'	—
TC-M1F1-04	Negative	Register with invalid email format	Email: user@ , @gmail.com, plaintext	1.Enter malformed email 2.Tap Register	Inline validation: 'Please enter a valid email address.'	—
TC-M1F1-05	Negative	Register with weak password	PW: 12345, password, abc123	1.Enter weak password 2.Tap	Error: 'Password must be 8+ chars with upper,	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
				Register	lower, digit & symbol.'	
TC-MiF1-06	Negative	Register with empty required fields	All fields blank	1.Open Sign Up 2.Tap Register without filling fields	All required fields highlighted red; submit blocked	—
TC-MiF1-07	UI/UX	Password strength indicator visible	Enter various passwords	1.Type weak password 2.Type medium 3.Type strong	Strength meter updates in real-time (Weak/Medium/Strong)	—
TC-MiF1-08	UI/UX	Form fields have correct keyboard types	Email, phone, password fields	1.Tap each field 2.Observe keyboard	Email->email keyboard; Phone->numeric; Password->default	—
TC-MiF1-09	Security	SQL injection in email field	Email: ' OR '1'='1	1.Enter SQL injection as email 2.Tap Register	Input sanitised; no DB error exposed; generic error shown	—
TC-MiF1-10	Performance	100 concurrent registrations	100 simultaneous register requests	1.Simulate 100 parallel register calls	All complete; response <3s; no 5xx errors	—

1.2 M1-F2: User Login (Email & Password)

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Email	registered@gmail.com	notregistered@x.com, null, empty	Exactly registered email	Case-insensitive match
Password	Correct password: Spice@2025	WrongPass, empty, spaces only	1 char wrong vs correct	Hash-compared server side
Session	Valid JWT token	Expired token, tampered token, null	Token 1 second before expiry	JWT HS256
Remember Me	Checked (persistent) / Unchecked	N/A	Session exactly at TTL	Cookie-based persistence

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-	Functional	Successful login with valid	Email:registered@gmail.com	1.Open Login 2.Enter email &	Login successful; JWT issued; Home	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
M1F2-01		credentials	PW:Spice@2025	password 3.Tap Login	screen shown	
TC-M1F2-02	Functional	Login with Remember Me checked	Valid credentials + Remember Me=on	1.Login 2.Enable Remember Me 3.Close & reopen app	User remains logged in without re-entering credentials	—
TC-M1F2-03	Functional	Successful logout clears session	Logged-in user	1.Go to Profile 2.Tap Logout	JWT invalidated; session cleared; user redirected to Login	—
TC-M1F2-04	Negative	Login with wrong password	Correct email, wrong password	1.Enter valid email 2.Enter wrong PW 3.Tap Login	Error: 'Incorrect password. Please try again.'	—
TC-M1F2-05	Negative	Login with unregistered email	Email: unknown@test.com	1.Enter unregistered email 2.Any password 3.Login	Error: 'No account found with this email.'	—
TC-M1F2-06	Negative	Account locked after 5 failed logins	5 wrong passwords consecutively	1.Enter wrong password 5 times	Account locked 15 min; 'Too many attempts' message shown	—
TC-M1F2-07	Negative	Login with empty fields	Both fields blank	1.Tap Login with empty fields	Both fields show red border; submit disabled	—
TC-M1F2-08	UI/UX	Password field masked with toggle	Any password input	1.Enter password 2.Observe field 3.Tap eye icon	Characters masked by default; eye icon toggles visibility	—
TC-M1F2-09	Security	Brute-force rate limiting	Rapid 50 login attempts/min	1.Send 50 requests/min to login API	Rate limiter triggers after 10 attempts; 429 Too Many Requests	—
TC-M1F2-10	Performance	Login response time on 4G	Single valid login, 4G network	1.Tap Login 2.Measure round-trip	Login completes in <2 seconds	—

1.3 M1-F3: OTP / Email Verification

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
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Category	Valid Data	Invalid Data	Boundary	Notes
OTP Code	123456 (valid 6-digit)	12345, abcdef, empty, 1234567	Exactly 6 digits	Numeric only
Expiry	OTP entered within 5 minutes	OTP entered after 5 min 1 sec	OTP at exactly 4:59 vs 5:01	Server-side expiry check
Resend	Request after 60 seconds	Rapid resend spam (<60s)	Exactly at 60-second cooldown	Cooldown enforced
Attempts	Max 3 wrong OTP attempts	4th wrong attempt	Exactly 3rd wrong attempt	OTP invalidated on 3 fails

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M1F3-01	Functional	Valid OTP submitted within expiry window	OTP: 123456, entered at 2 min	1.Register 2.Receive OTP 3.Enter correct OTP within 5 min	Email verified; account activated; redirect to app	—
TC-M1F3-02	Functional	Resend OTP after 60-second cooldown	Wait 61 seconds after first OTP	1.Register 2.Wait 61 sec 3.Tap Resend OTP	New OTP sent; old OTP invalidated; 60s timer resets	—
TC-M1F3-03	Functional	OTP screen shows countdown timer	Just after OTP sent	1.Open OTP screen 2.Observe timer	Countdown from 5:00 visible; Resend button disabled until 0	—
TC-M1F3-04	Negative	OTP entered after 5-minute expiry	OTP entered at 5 min 30 sec	1.Register 2.Wait 6 min 3.Enter OTP	Error: 'OTP has expired. Please request a new one.'	—
TC-M1F3-05	Negative	Wrong OTP entered 3 times	3 consecutive wrong OTPs	1.Enter wrong OTP 3 times	After 3rd wrong attempt: 'OTP invalidated. Request a new code.'	—
TC-M1F3-06	Negative	Resend OTP before 60s cooldown	Tap Resend at 30 seconds	1.Tap Resend before cooldown expires	Resend button disabled; timer shown; no new OTP sent	—
TC-M1F3-07	Negative	Invalid OTP format (non-numeric)	OTP: abcdef, 12.34, spaces	1.Enter alphabetic/non-numeric OTP	Error: 'Please enter a valid 6-digit numeric code.'	—
TC-M1F3-08	UI/UX	OTP input auto-advances between digits	6-digit OTP entry	1.Enter digit 1 2.Observe cursor	Cursor auto-moves to next digit box after each entry	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M1F3-09	Security	OTP not reusable after successful use	Same OTP reused	1.Verify with OTP 2.Re-enter same OTP	Error: 'This OTP has already been used.'	—
TC-M1F3-10	Security	OTP not guessable (random, not sequential)	Observe 5 consecutive OTPs	1.Register 5 times 2.Compare OTPs	OTPs are cryptographically random; no sequential pattern	—

1.4 M1-F4: Social Login (Google / Facebook OAuth)

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Google Token	Valid, non-expired Google OAuth token	Revoked, expired, wrong-provider token	Token 1 sec before expiry	OAuth 2.0 PKCE flow
Facebook Token	Valid Facebook access token	Expired FB token, cancelled permission	Token at expiry boundary	FB Login SDK
Profile Data	Name, email, profile photo from OAuth	Missing email in OAuth scope	Name = max 255 chars	Email required from provider
Account Merge	Social login email matches existing	Conflicting email, duplicate account	First-time vs returning user	Merge prompt shown

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M1F4-01	Functional	First-time Google login creates account	Valid Google account token	1.Tap Sign in with Google 2.Select account 3.Grant permission	Account auto-created; Google profile imported; Welcome screen	—
TC-M1F4-02	Functional	Returning user logs in via Google	Same Google account used before	1.Tap Sign in with Google 2.Select same account	Existing account opened; no duplicate created	—
TC-M1F4-03	Functional	Facebook OAuth creates account	Valid Facebook account	1.Tap Sign in with Facebook 2.Authorize	Account created; FB profile photo shown	—
TC-M1F4-04	Functional	Social login email matches existing email account — merge prompt	Google email = existing SpiceSync email	1.Register with email 2.Later sign in with Google (same email)	Prompt: 'Account exists. Link Google to this account?' Merge offered	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M1F4-05	Negative	Revoked Google OAuth token	Previously valid but revoked token	1.Use revoked Google token for login	Error: 'Authentication failed. Please try signing in again.'	—
TC-M1F4-06	Negative	Facebook login with missing email scope	Facebook account without email permission	1.Sign in with Facebook 2.Deny email permission	Error: 'Email access required. Please grant permission to continue.'	—
TC-M1F4-07	Negative	Expired OAuth token	Token expired by >1 hour	1.Use expired token	Token rejected; user redirected to re-authenticate	—
TC-M1F4-08	UI/UX	Social login buttons clearly branded	Sign in screen	1.View login screen	Google button shows Google logo; Facebook shows FB logo; correct brand colours	—
TC-M1F4-09	Security	OAuth state parameter validated (CSRF)	Intercept OAuth callback	1.Modify state param in OAuth callback URL	State mismatch detected; login rejected; CSRF attack blocked	—
TC-M1F4-10	Security	OAuth token not stored in plain text	Successful OAuth login	1.Complete OAuth 2.Inspect app storage/logs	Token stored as encrypted value; never in plain-text logs	—

1.5 M1-F5: Forgot Password / Password Reset

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Email Input	Registered email: user@gmail.com	Unregistered email, null, invalid format	Exactly registered email	Email sent only if registered
Reset Link	Fresh link within 30-min expiry	Link used twice, expired link	Link at exactly 29:59 vs 30:01	One-time-use token
New Password	Valid: NewSpice@99	Same as old, weak, empty	8 chars minimum	Old password cannot be reused
Token	Unique reset token in URL	Tampered/guessed token	N/A	Cryptographically random

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M1F5-01	Functional	Request password reset — registered email	Email: user@gmail.com	1.Tap Forgot Password 2.Enter registered email 3.Tap Send Reset Link	Reset email sent within 2 min; success message shown	—
TC-M1F5-02	Functional	Follow reset link and set new password	Valid link, new PW: NewSpice@99	1.Click link in email 2.Enter new password 3.Confirm 4.Submit	Password changed; login with new password succeeds	—
TC-M1F5-03	Functional	Old password no longer works after reset	Old PW: Spice@2025	1.Reset password 2.Attempt login with old password	Error: 'Incorrect password.' Old password rejected	—
TC-M1F5-04	Negative	Reset request with unregistered email	Email: unknown@mail.com	1.Enter unregistered email 2.Tap Send	Generic message: 'If this email is registered, a link has been sent.' No email sent	—
TC-M1F5-05	Negative	Reset link used twice	Same reset link reused	1.Reset password via link 2.Use same link again	Error: 'This reset link has already been used or has expired.'	—
TC-M1F5-06	Negative	Reset link expired (>30 min)	Link accessed after 35 minutes	1.Request reset 2.Wait 35 min 3.Click link	Error: 'This reset link has expired. Please request a new one.'	—
TC-M1F5-07	Negative	New password same as old password	New PW = Old PW	1.Enter new password identical to old 2.Submit	Error: 'New password cannot be the same as your current password.'	—
TC-M1F5-08	UI/UX	Reset email sent confirmation message visible	After submitting reset form	1.Enter email 2.Tap Send	Clear confirmation message shown; no ambiguity about next step	—
TC-M1F5-09	Security	Reset token is unguessable	Observe 5 reset tokens	1.Request 5 resets 2.Compare token structure	Tokens are UUIDs or 256-bit random strings; no pattern	—
TC-M1F5-10	Security	Reset token invalidated after use and on new request	Request 2 resets, use second	1.Request reset (Token A) 2.Request again (Token B) 3.Use Token A	Token A rejected; only Token B is valid	—

1.6 M1-F6: Profile Management

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Name	Ali Khan, Fatima Noor	Empty, >100 chars, injection	100 chars (max), 2 chars (min)	Unicode / Urdu names supported
Profile Photo	JPEG/PNG <5MB	>10MB, .exe, .pdf, null	Exactly 5MB file	Images only; resized to 200x200
Phone Update	+92-300-9999999	Landline, <10 digits, letters	Exactly 10-digit local	Must re-verify on change
Email Update	new@gmail.com	Invalid format, already used	64-char local part	OTP re-verification required

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M1F6-01	Functional	Update display name successfully	New name: Ahmed Raza	1.Open Profile 2.Edit name 3.Enter Ahmed Raza 4.Save	Name updated; reflected immediately in header and profile	—
TC-M1F6-02	Functional	Upload valid profile photo	JPEG file, 2MB	1.Tap profile photo 2.Select 2MB JPEG 3.Confirm	Photo uploaded; resized to display; old photo replaced	—
TC-M1F6-03	Functional	Update phone number with OTP verification	New: +92-321-8888888	1.Edit phone 2.Enter new number 3.Verify OTP	Phone updated after OTP confirmation	—
TC-M1F6-04	Functional	Switch language preference Urdu <-> English	Language toggle in profile	1.Open Profile Settings 2.Toggle language to Urdu	Entire app switches to Urdu; persists across sessions	—
TC-M1F6-05	Negative	Upload photo exceeding 5MB	File: 8MB PNG	1.Select 8MB photo 2.Tap Upload	Error: 'Photo must be under 5MB. Please choose a smaller image.'	—
TC-M1F6-06	Negative	Upload non-image file as profile photo	File: resume.pdf	1.Attempt to select PDF as photo	Error: 'Only JPEG, PNG, and WebP images are allowed.'	—
TC-M1F6-07	Negative	Save profile with empty name	Name field cleared	1.Clear name field 2.Tap Save	Error: 'Display name cannot be empty.'	—
TC-M1F6-08	UI/UX	Profile photo preview before upload	Selecting a photo	1.Select photo 2.Observe preview before confirming	Crop/preview screen shown; user can confirm or retake	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M1F6-09	Security	XSS in name field	Name: <script>alert(1)</script>	1.Enter XSS payload as name 2.Save	Input sanitised; script tag stripped; safe text stored	—
TC-M1F6-10	Security	Email change requires re-authentication	Change email to new@gmail.com	1.Edit email 2.Enter new email 3.Confirm with OTP on old email	Old email OTP required before new email is activated	—

1.7 M1-F7: Account Deletion

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Confirmation	User types 'DELETE' to confirm	Types 'delete', 'Delete', wrong word	Exactly 'DELETE' (case-sensitive)	Hard delete or soft-delete flag
Password Check	Correct current password	Wrong password before deletion	N/A	Re-auth required
Active Orders	No active orders at time of deletion	1 active order in transit	Order delivered 1 min before delete	Cannot delete with pending orders
Data Retention	Account data purged per GDPR/PDPA	N/A	Data at exactly 30-day grace period	Anonymisation of past orders

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M1F7-01	Functional	Account deletion with no active orders	Valid password, type DELETE, no orders	1.Go to Settings 2.Delete Account 3.Re-enter password 4.Type DELETE 5.Confirm	Account deleted; session ended; login with this email fails	—
TC-M1F7-02	Functional	Account data cleared from public view post-deletion	Deleted account	1.Delete account 2.Search username in public areas	Username, profile photo, reviews no longer visible publicly	—
TC-M1F7-03	Negative	Delete account with active order in transit	1 order with status 'On the Way'	1.Attempt account deletion	Error: 'You have an active order. Please wait for delivery before deleting.'	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M1F7-04	Negative	Delete confirmation with wrong keyword	Type 'delete' (lowercase) instead of 'DELETE'	1.Type delete (lowercase) 2.Tap Confirm	Delete button remains disabled; hint shown: 'Type DELETE in uppercase.'	—
TC-M1F7-05	Negative	Delete with wrong password confirmation	Wrong current password entered	1.Enter wrong password in re-auth step	Error: 'Incorrect password. Account not deleted.'	—
TC-M1F7-06	Negative	Re-register with deleted account email	Email of deleted account	1.Delete account 2.Try to register same email	Same email can be re-registered after 24-hour cool-off period	—
TC-M1F7-07	UI/UX	Multiple confirmation steps before deletion	Deletion flow	1.Navigate to Delete Account	At least 2 confirmation screens shown; warning about permanence displayed	—
TC-M1F7-08	UI/UX	Delete button is red / visually distinct warning	Account deletion page	1.Open Delete Account page	Delete button shown in red; destructive action clearly indicated	—
TC-M1F7-09	Security	Session tokens invalidated immediately on deletion	Active sessions on multiple devices	1.Delete account 2.Try to use old JWT from another device	JWT rejected; 401 Unauthorized on all devices	—
TC-M1F7-10	Security	Personal data purged within 30 days (PDPA/GDPR)	Deleted account, 30-day review	1.Delete account 2.After 30 days admin checks DB	PII fields (name, email, phone) anonymised or deleted from DB	—

1.8 M1-F8: Notification Preferences

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Notification Types	Order Updates, Promotions, New Menu, Loyalty Points	Unknown type, null	All enabled vs all disabled	Per-channel toggle
Channels	Push, Email, SMS	Invalid token, unsubscribed SMS	Exactly 1 channel active	Can disable all
Quiet Hours	Set: 10 PM – 7 AM	End time before start time	Notification at exactly 10:00 PM	Timezone-aware

Category	Valid Data	Invalid Data	Boundary	Notes
Language	Urdu, English	Unsupported language code	Toggle mid-session	Applies to all notifications

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M1F8-01	Functional	Disable promotional notifications	Promos toggle set to OFF	1.Settings > Notifications 2.Turn off Promotions 3.Save	No promotional push/email received; order updates unaffected	—
TC-M1F8-02	Functional	Enable all notification channels	All channels set to ON	1.Toggle Push, Email, SMS all ON 2.Save	User receives notifications on all 3 channels for order updates	—
TC-M1F8-03	Functional	Set quiet hours 10 PM – 7 AM	Quiet hours configured	1.Set quiet start 10 PM 2.End 7 AM 3.Save 4.Trigger notification at 11 PM	Notification queued; delivered at 7:01 AM next day	—
TC-M1F8-04	Functional	Switch notification language to Urdu	Language: Urdu in notification settings	1.Set notification language to Urdu	All push and email notifications arrive in Urdu script	—
TC-M1F8-05	Negative	Set quiet hours end time before start	Start: 10 PM, End: 6 PM (same day)	1.Enter end time earlier than start time 2.Save	Error: 'End time must be after start time.'	—
TC-M1F8-06	Negative	Disable all notifications then place order	All notifications OFF	1.Disable all channels 2.Place an order	No push/email/SMS received for order — system respects preference	—
TC-M1F8-07	UI/UX	Notification preference screen is scannable	Settings > Notifications screen	1.Open Notifications settings	Toggles grouped by type; clear ON/OFF state; no horizontal scroll	—
TC-M1F8-08	UI/UX	Quiet hours time picker is intuitive	Setting quiet hours	1.Tap quiet hours time fields	Native time picker (clock wheel or spinner) used; AM/PM clear	—
TC-M1F8-09	Security	Push notification does not contain sensitive data	Order confirmed notification	1.Receive push notification 2.Inspect payload in notification tray	Notification shows generic text (e.g., 'Your order is confirmed!'); no payment data or PII in payload	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M1F8-10	Performance	Notification delivered within 30 seconds of trigger	Order status changed	1.Change order status in backend 2.Measure delivery time to device	Push notification received on device within 30 seconds	—

1.9 M1-F9: Session Management

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Session Token	Valid JWT, not expired	Expired JWT, tampered signature	Token at exactly TTL – 1 second	HS256 signed
Concurrent Sessions	2 devices logged in simultaneously	N/A	Exactly 3 concurrent sessions	Max 3 concurrent sessions allowed
Idle Timeout	Active use	Idle for 30 min	Action at exactly 29:59	Auto-logout on idle
Forced Logout	Admin forces logout	N/A	N/A	All tokens invalidated

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M1F9-01	Functional	Session persists on app reopen (within TTL)	Valid session, app closed and reopened	1.Login 2.Close app 3.Reopen within 1 hour	User still logged in; no re-login required	—
TC-M1F9-02	Functional	Session expires after idle timeout (30 min)	Idle app for 31 minutes	1.Login 2.Do nothing for 31 min 3.Attempt action	Session expired; user redirected to Login screen	—
TC-M1F9-03	Functional	Simultaneous login on 2 devices	Same account on phone A and phone B	1.Login on phone A 2.Login on phone B 3.Perform action on both	Both sessions active up to max 3; each gets own JWT	—
TC-M1F9-04	Functional	Token auto-refreshes silently near expiry	Token near TTL, user is active	1.Use app near token expiry 2.Perform action	Token silently refreshed; no login prompt shown to user	—
TC-M1F9-05	Negative	Tampered JWT rejected	Modified JWT payload	1.Decode JWT 2.Change user_id 3.Re-encode 4.Send request	Server rejects tampered token; 401 Unauthorised	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M1F9-06	Negative	Expired JWT rejected	JWT expired by 1 minute	1.Use JWT 1 minute after expiry	Server returns 401; client redirected to Login	—
TC-M1F9-07	Negative	Exceed max concurrent sessions (4th device)	Login on 4th device	1.Login on 3 devices 2.Login on 4th	Oldest session invalidated; user on device 1 logged out	—
TC-M1F9-08	UI/UX	Session expiry shows friendly message	Session expired during app use	1.Let session expire 2.Tap any button	Dialog: 'Your session has expired. Please log in again.' — not a crash	—
TC-M1F9-09	Security	JWT not stored in insecure local storage	Login flow inspection	1.Login 2.Inspect app storage (via ADB or dev tools)	JWT stored in encrypted Keychain (iOS) / Keystore (Android), not plain localStorage	—
TC-M1F9-10	Security	Logout invalidates token server-side	Token captured before logout, used after	1.Capture JWT 2.Logout 3.Use captured JWT in API call	Token blacklisted; 401 returned even with previously valid JWT	—

1.10 M1-F10: Urdu Language Support (Account Screens)

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Language Toggle	English, Urdu	Invalid locale code, null	Mid-session toggle	RTL layout for Urdu
Urdu Text Fields	Urdu name input in registration	Mixing Urdu + ASCII in same field	Max-length Urdu string (30 chars)	Nastaliq or Naskh font
Error Messages	All errors in Urdu	English fallback if Urdu string missing	N/A	All 50+ error strings translated
Font Rendering	Nastaliq font renders correctly	Missing font fallback	N/A	Embedded font required

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-	Functional	All registration	Language set to	1.Set language Urdu	All field labels,	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
M1F10-01		labels appear in Urdu	Urdu before opening Sign Up	2.Open Sign Up form	placeholders, button text in Urdu script	
TC-M1F10-02	Functional	Login screen switches to Urdu	Language toggle: Urdu	1.Toggle Urdu on Login screen	Login, email, password, forgot password all in Urdu	—
TC-M1F10-03	Functional	Urdu name stored and displayed correctly	Name entered in Urdu:	1.Register with Urdu name 2.View profile	Urdu name stored correctly; displayed with proper rendering	—
TC-M1F10-04	Functional	Language preference persists after app restart	Urdu selected, app restarted	1.Set Urdu 2.Close app 3.Reopen	App opens in Urdu; preference saved across sessions	—
TC-M1F10-05	Functional	Error messages shown in Urdu	Urdu mode + trigger validation errors	1.In Urdu mode 2.Submit blank registration form	All validation errors displayed in Urdu script	—
TC-M1F10-06	Negative	Switch to unsupported language code	Locale: 'xx' (non-existent)	1.Force locale xx via settings	App falls back to English; no crash; error logged	—
TC-M1F10-07	Negative	Mixed-direction input (Urdu + English) in name	Name: Ali	1.Enter mixed name 2.Save	Mixed text saved and displayed correctly using BiDi rendering	—
TC-M1F10-08	UI/UX	Layout mirrors for RTL when Urdu active	Urdu mode on registration	1.Set Urdu 2.Open any account form	Form mirrors to RTL; labels on right; inputs right-aligned	—
TC-M1F10-09	UI/UX	Urdu font renders clearly on all screen sizes	Urdu mode on different devices	1.Open app in Urdu on 4" and 6.5" screens	Nastaliq/Naskh font legible; no clipping or overlap	—
TC-M1F10-10	Performance	Language switch completes in <1 second	Toggle English -> Urdu	1.Tap language toggle 2.Measure UI re-render time	Full UI re-renders in under 1 second; no flash/blank screen	—

2. Module M2: Order Placement & Cart Management

Covers the complete food ordering journey: browsing the menu, adding items with customisations (spice levels, region), cart management, promo codes, group orders, checkout, and order confirmation.

2.1 M2-F1: Browse Menu & Item Discovery

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Category Filter	Burgers, Biryani, Pizza, Deals, Beverages	Invalid category ID, null	All categories vs single	5 categories available
Search Query	'Burger', 'Biryani', 'Peri Peri'	Empty string, numeric, >100 chars	1-char query, exact item name	Case-insensitive search
Sort Options	Price (low-high), Popularity, New	Null sort option	Price = PKR o item	3 sort modes
City Filter	Islamabad, Karachi, Lahore, Multan, Abbottabad	City not in list, null	Single city with 1 menu item	Menu varies per city

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M2F1-01	Functional	Filter menu by 'Burgers' category	Category: Burgers	1.Open Menu 2.Tap Burgers filter	Only burger items displayed; count badge correct	—
TC-M2F1-02	Functional	Search for 'Biryani' returns correct results	Query: Biryani	1.Tap search bar 2.Type Biryani	All biryani items shown; non-biryani items hidden	—
TC-M2F1-03	Functional	Sort items by Price (Low to High)	Sort: Price ascending	1.Open any category 2.Select sort: Price Low-High	Items reordered by price ascending; PKR values correct	—
TC-M2F1-04	Functional	City-specific menu loads for Karachi	City: Karachi	1.Set delivery city to Karachi 2.Browse menu	Only menu items available in Karachi shown	—
TC-M2F1-05	Negative	Search with empty string	Query: empty	1.Tap search 2.Leave empty 3.Tap Search	Full menu shown or prompt: 'Enter a search term.'	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M2F1-06	Negative	Filter by unavailable city	City: Quetta (not supported)	1.Select Quetta in city dropdown	Error: 'SpiceSync is not available in Quetta yet.'	—
TC-M2F1-07	Negative	Search returns no results	Query: 'zzz_nothing'	1.Search for non-existent item	Empty state shown: 'No items found. Try a different search.'	—
TC-M2F1-08	UI/UX	Menu item cards show price, name, photo, rating	Any menu category	1.Open Burgers category 2.Inspect item cards	Each card: photo, name, price in PKR, star rating, Add button visible	—
TC-M2F1-09	UI/UX	Sold-out items visually distinct	Item marked unavailable	1.Browse menu with sold-out items	Sold-out items show grey overlay + 'Sold Out' badge; Add button disabled	—
TC-M2F1-10	Performance	Menu loads in <3 sec with 50+ items	Full menu, standard 4G	1.Open Menu tab 2.Measure load time	All items visible within 3 seconds; images lazy-load	—

2.2 M2-F2: Add to Cart & Quantity Management

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Item	Beef Burger PKR 450, Classic Pizza PKR 750	Non-existent item ID 99999, deleted item	Cheapest and most expensive item	Price fetched server-side
Quantity	1, 5, 10, 20	0, -1, 21, 999, 'five' (text)	Exactly 20 (max), 1 (min)	Integer validation
Cart Total	PKR 100 to PKR 49,999	PKR 0, PKR 50,001	PKR 50,000 (cap)	Calculated server-side
Item Options	Spice level, size (regular/large)	Invalid option ID, null	All options at maximum	Options add price delta

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M2F2-01	Functional	Add single item to cart	Beef Burger x1	1.Open item 2.Tap Add to Cart	Cart badge shows 1; subtotal = PKR 450	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M2F2-02	Functional	Add multiple different items	Beef Burger, Pizza, Fries	1.Add 3 different items	Cart contains 3 items; total = sum of prices	—
TC-M2F2-03	Functional	Increase item quantity in cart	Beef Burger 1->3	1.Open cart 2.Tap + on Burger twice	Quantity = 3; subtotal updated in real-time	—
TC-M2F2-04	Functional	Decrease item quantity in cart	Burger qty 3->2	1.Tap – on Burger in cart	Quantity = 2; subtotal reduced	—
TC-M2F2-05	Functional	Remove item from cart	Burger in cart, tap remove	1.Swipe item left in cart 2.Tap Remove	Item removed; total updated; if last item, empty cart screen shown	—
TC-M2F2-06	Negative	Add item with quantity 0	Qty field set to 0	1.Open item 2.Set qty to 0 3.Tap Add	Error: 'Quantity must be at least 1.' Item not added.	—
TC-M2F2-07	Negative	Exceed max quantity per item (>20)	Qty: 21	1.Set quantity to 21	Error: 'Maximum 20 units per item allowed.'	—
TC-M2F2-08	Negative	Add deleted/unavailable item via deep link	Item ID that was removed from menu	1.Use old deep link to add deleted item	Error: 'This item is no longer available.' Item not added.	—
TC-M2F2-09	UI/UX	Cart badge updates instantly on add	Add item to cart	1.Tap Add to Cart 2.Observe nav bar	Cart badge increments immediately; no page reload needed	—
TC-M2F2-10	Performance	Add to cart response time	Single add operation on 4G	1.Tap Add to Cart 2.Measure time to cart update	Cart updates in <1 second	—

2.3 M2-F3: Spice Level Customisation

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Spice Level	Mild, Medium, Hot, Peshawari, Punjabi, Sindhi	Invalid string, null, numeric 7	All 6 predefined levels	Enum; server validates
Item Support	Biryani, Karahi, Nihari, Curries	Beverages, desserts (no spice option)	Item with all 6 options	Not all items have spice option

Category	Valid Data	Invalid Data	Boundary	Notes
Price Delta	Mild=+PKR 0, Peshawari=+PKR 30	Negative price delta	Free (PKR 0) vs paid (+PKR 50)	Delta added to base price
Persistence	Spice saved in cart	Cart cleared on session end (no login)	Same item 2x with different spice	Each cart entry independent

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M2F3-01	Functional	Select Peshawari spice for Biryani	Item: Biryani, Spice: Peshawari	1.Open Biryani 2.Select Peshawari spice 3.Add to cart	Item added with Peshawari label; +PKR 30 delta applied	—
TC-M2F3-02	Functional	Add same item twice with different spice levels	Biryani x Mild + Biryani x Sindhi	1.Add Biryani (Mild) 2.Add Biryani again (Sindhi)	Cart shows 2 separate entries; each with correct spice and price	—
TC-M2F3-03	Functional	Spice option not available for beverages	Item: Mango Shake (no spice)	1.Open Mango Shake item detail	Spice level selector not shown; only size/extras options	—
TC-M2F3-04	Functional	Default spice level pre-selected	Open any curry item	1.Open Karahi item	Medium spice pre-selected by default; user can change	—
TC-M2F3-05	Negative	Submit order with null spice (API tamper)	Force null spice in API payload	1.Intercept add-to-cart request 2.Set spice=null 3.Submit	Server rejects; returns 400 Bad Request: 'spice_level is required'	—
TC-M2F3-06	Negative	Submit invalid spice level string (API tamper)	spice_level: 'ULTRA_HOT'	1.Intercept request 2.Set spice to invalid string	Server rejects; 422 Unprocessable Entity; enum validation fails	—
TC-M2F3-07	UI/UX	Spice levels colour-coded (green to red)	Spice selector on any curry item	1.Open item with spice options 2.View selector	Mild=green, Medium=yellow, Hot=orange, Peshawari/Punjabi/Sindhi=red gradient	—
TC-M2F3-08	UI/UX	Spice level shown on cart item summary	Biryani with Peshawari in cart	1.Add item with spice 2.Open cart	Cart line item shows item name + spice level label below it	—
TC-M2F3-09	UI/UX	Urdu spice labels visible when language is Urdu	Urdu mode, spice selector	1.Set Urdu 2.Open item with spice	Spice names displayed in Urdu (, , , etc.)	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M2F3-10	Security	Spice delta price validated server-side	Client modifies price delta to PKR 0	1.Intercept request 2.Set spice_price_delta=0 for Peshawari	Server ignores client price; uses server-defined PKR 30 delta	—

2.4 M2-F4: Promo Codes & Discounts

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Promo Code	EID2025 (10%), RAMADAN20 (20%), FIRST50 (PKR 50 off)	EXPIREDCODE, XXXXX, empty, used code	Code at exactly discount cap	Case-insensitive; single-use per account
Min Order	PKR 500+ for EID2025	PKR 499 (below minimum)	Exactly PKR 500	Minimum order enforced
Discount Type	Percentage (10%), Fixed (PKR 50)	N/A	100% discount edge case	Both types supported
Stacking	1 promo code per order	2 promo codes simultaneously	Promo + loyalty points (allowed)	Only 1 promo at a time

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M2F4-01	Functional	Apply valid percentage promo code	Code: EID2025 (10% off), Order PKR 800	1.Go to cart 2.Enter EID2025 3.Tap Apply	10% off = PKR 80 discount; new total PKR 720 shown	—
TC-M2F4-02	Functional	Apply fixed-amount promo code	Code: FIRST50 (PKR 50 off), Order PKR 500	1.Enter FIRST50 2.Apply	PKR 50 deducted; total = PKR 450	—
TC-M2F4-03	Functional	Remove applied promo code	Code applied, tap Remove	1.Apply promo 2.Tap X / Remove code	Discount removed; original total restored	—
TC-M2F4-04	Functional	Promo code applied + Loyalty Points stacked	EID2025 + 200 Spice Points	1.Apply EID2025 2.Redem 200 points	Both discounts applied; total correctly reduced by both amounts	—
TC-M2F4-05	Negative	Apply expired promo code	Code: EXPIREDCODE	1.Enter EXPIREDCODE 2.Tap Apply	Error: 'This promo code has expired or	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
					does not exist.'	
TC-M2F4-06	Negative	Apply promo on order below minimum	Code: EID2025 (min PKR 500), Order PKR 300	1.Add PKR 300 item 2.Enter EID2025	Error: 'Minimum order of PKR 500 required for this code.'	—
TC-M2F4-07	Negative	Apply already-used single-use promo	Code: FIRST50 (already used by this account)	1.Use FIRST50 on 2nd order	Error: 'This promo code has already been used.'	—
TC-M2F4-08	Negative	Apply 2 promo codes simultaneously	EID2025 + RAMADAN20	1.Apply first code 2.Enter second code	Error: 'Only one promo code can be applied per order.'	—
TC-M2F4-09	UI/UX	Savings line clearly shown on cart	Promo applied	1.Apply promo 2.View cart total section	Breakdown shows: Subtotal, Discount (- PKR X), Total — clearly labelled	—
TC-M2F4-10	Security	Promo discount validated server-side	Client modifies discount amount	1.Apply promo 2.Intercept checkout request 3.Inflate discount	Server ignores client discount; uses server-calculated value	—

2.5 M2-F5: Delivery Address & Location Selection

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
City	Islamabad, Karachi, Lahore, Multan, Abbottabad	Quetta, 'Mars', null, numeric	All 5 supported cities	5 cities supported
Area	E-11 ISB, Clifton KHI, Gulberg LHR	Area outside supported zones	Furthest zone in each city	Dependent on city
GPS Auto-detect	User location within delivery zone	Location outside Pakistan, denied	On boundary of delivery zone	Requires location permission
Saved Address	Up to 5 saved addresses	6th address (exceeds limit)	Exactly 5 saved addresses	Max 5 per account

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
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TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M2F5-01	Functional	Select city (Islamabad) and area (E-11)	City: Islamabad, Area: E-11	1.Go to Delivery 2.Select Islamabad 3.Select E-11	Area confirmed; delivery fee calculated; ETA shown	—
TC-M2F5-02	Functional	Auto-detect current location	GPS on, user in Lahore	1.Tap Use My Location 2.Grant permission	GPS detects Lahore; nearest delivery zone auto-selected	—
TC-M2F5-03	Functional	Save a new address	Home address in Karachi	1.Enter delivery address 2.Tap Save Address 3.Name it 'Home'	Address saved; appears in saved addresses list	—
TC-M2F5-04	Functional	Select previously saved address	Saved address: Home	1.Open saved addresses 2.Select Home	Home address auto-populated in delivery details	—
TC-M2F5-05	Negative	Select delivery area outside supported zones	Area: remote village not in list	1.Type unsupported area name	Error: 'Delivery is not available in this area yet.'	—
TC-M2F5-06	Negative	Location permission denied — GPS unavailable	GPS permission denied	1.Tap Use My Location 2.Deny permission	Fallback to manual city/area selection; no crash	—
TC-M2F5-07	Negative	Add 6th saved address (exceeds limit)	Already 5 saved addresses	1.Try to add 6th address	Error: 'Maximum 5 addresses allowed. Delete one to add new.'	—
TC-M2F5-08	Negative	Proceed to checkout without selecting address	No address selected	1.Tap Checkout 2.Skip address	Error: 'Please select a delivery address before placing your order.'	—
TC-M2F5-09	UI/UX	Area dropdown filters based on selected city	City selected: Lahore	1.Select Lahore 2.Open Area dropdown	Only Lahore delivery areas shown; other city areas hidden	—
TC-M2F5-10	UI/UX	Delivery fee prominently shown before checkout	Delivery to E-11 Islamabad	1.Select address 2.View order summary	Delivery fee line visible in summary; not hidden until payment	—

2.6 M2-F6: Group Order Feature

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
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Category	Valid Data	Invalid Data	Boundary	Notes
Group Size	2–100 participants	1 (no group), 0, -1, 101	Exactly 2 (min), 100 (max)	Integer, invite-link based
Invite Link	Valid shareable link	Expired link (>30 min), used link	Link at exactly 30-min expiry	Link expires after 30 min or order placed
Item Addition	Each member adds their own items	Member adds after host confirms	All 100 members add simultaneously	Merge at checkout
Discount	10% group discount for 3+ members	Group of 2 (no discount)	Exactly 3 members (discount kicks in)	Discount auto-applied

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M2F6-01	Functional	Host creates group order and shares link	Logged-in user creates group	1.Tap Group Order 2.Share generated link	Unique invite link generated; shareable via WhatsApp/SMS	—
TC-M2F6-02	Functional	3 members join and add items	3 users join via link	1.Host creates 2.2 others join via link 3.Each adds items	All 3 carts merged; 10% group discount applied	—
TC-M2F6-03	Functional	Host confirms and places group order	All members have added items	1.Host reviews merged cart 2.Confirm 3.Pays	Single order placed; all members receive order confirmation	—
TC-M2F6-04	Functional	Group member removes own item	Member A removes their item	1.Member A opens own cart portion 2.Removes item	Item removed from merged cart; total updated; others unaffected	—
TC-M2F6-05	Negative	Join group order with expired invite link	Link expired after 35 minutes	1.Click link after 35 min	Error: 'This group order link has expired.'	—
TC-M2F6-06	Negative	Non-member attempts to modify another's cart	User B modifies User A's items	1.User B intercepts User A's cart API call 2.Modifies item	403 Forbidden; cart modification rejected	—
TC-M2F6-07	Negative	Group order placed after member goes offline	1 of 3 members goes offline	1.Member 3 loses connection 2.Host confirms order	Order placed with available member carts; offline member's items included if already	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
					added	
TC-M2F6-o8	UI/UX	Group order screen shows all member names and items	Active group order	1.Open group cart view	Each member's name shown as section header; their items listed below	—
TC-M2F6-o9	UI/UX	Countdown timer visible on group order screen	Group order created	1.Open group cart 2.Observe timer	30-minute countdown visible; flashes red in last 5 minutes	—
TC-M2F6-10	Performance	100-member group order merge and checkout	100 participants add items simultaneously	1.Simulate 100 concurrent item adds to group	Merge completes in <5 sec; no cart corruption; correct total	—

2.7 M2-F7: Reorder from History

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Order History	Past orders with status 'Delivered'	Cancelled/failed orders	Most recent order (1 min ago)	Last 30 orders shown
Item Availability	All items still on menu	1 item removed from menu	Item price changed since last order	Availability checked at reorder
Customisation	Same spice level as original	Spice option no longer available	N/A	Re-applies saved customisation
Cart State	Cart empty before reorder	Existing items in cart	Cart at 20-item max before reorder	Merge or replace prompt

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M2F7-01	Functional	Reorder identical past order	Delivered order with all items available	1.Order History 2.Select past order 3.Tap Reorder	All items added to cart with same customisations	—
TC-	Functional	Reorder when item	Price of Biryani increased since	1.Tap Reorder 2.Observe	New price applied; toast: 'Prices may	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
M2F7-02		price has changed	last order	cart	have changed since your last order.'	
TC-M2F7-03	Functional	Reorder when cart has existing items — merge prompt	Cart has items, user taps Reorder	1.Add item to cart 2.Tap Reorder from history	Prompt: 'Replace cart or add to existing?' — both options available	—
TC-M2F7-04	Functional	Reorder from 30-day-old order	Order from 30 days ago	1.Scroll to old order 2.Tap Reorder	Items added if available; unavailable items flagged	—
TC-M2F7-05	Negative	Reorder when one item no longer on menu	Deleted item in past order	1.Tap Reorder 2.Item no longer in menu	Toast: 'Peri Peri Burger is no longer available and was skipped.' Other items added.	—
TC-M2F7-06	Negative	Reorder cancelled/failed past order	Cancelled order in history	1.Open cancelled order	Reorder button still present; items re-added normally if available	—
TC-M2F7-07	Negative	Reorder when cart is at max (20 items)	Cart has 20 items	1.Tap Reorder (5-item past order)	Error or prompt: 'Cart is full. Remove items to reorder.'	—
TC-M2F7-08	UI/UX	Reorder button clearly visible in Order History	Order History screen	1.Open Order History 2.Inspect each order card	Each delivered order card has prominent 'Reorder' button	—
TC-M2F7-09	UI/UX	Order history shows date, items, and total	Past orders list	1.Open Order History	Each entry shows: date, restaurant, item count, total in PKR	—
TC-M2F7-10	Performance	Reorder adds to cart in <1 second	Reorder any past order	1.Tap Reorder 2.Measure cart update time	Cart updated in under 1 second	—

2.8 M2-F8: Checkout & Order Confirmation

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Cart State	1–20 items, valid delivery address, payment ready	Empty cart, no address, no payment	Single item order	All three required
Delivery Type	Delivery, Pickup	N/A	Delivery fee =	Two modes

Category	Valid Data	Invalid Data	Boundary	Notes
			PKR 0 (promo)	
Order Total	PKR 100 – PKR 49,999	PKR 0, negative, PKR 50,001	PKR 50,000 (max cart value)	Server-validated
Confirmation	Order ID: UUID v4	Duplicate order submission	N/A	Idempotency key used

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M2F8-01	Functional	Successful checkout — delivery order	Valid cart, Islamabad address, JazzCash	1.Add items 2.Select delivery 3.Pay via JazzCash 4.Confirm	Order confirmed; Order ID generated; confirmation screen shown	—
TC-M2F8-02	Functional	Successful checkout — pickup order	Valid cart, Lahore pickup branch	1.Add items 2.Choose Pickup: Gulberg branch 3.Pay 4.Confirm	Order placed; pickup address and ready-time shown	—
TC-M2F8-03	Functional	Order confirmation email sent	Any successful order	1.Complete checkout	Confirmation email with order details arrives within 2 minutes	—
TC-M2F8-04	Functional	Order ID is unique per order	Place 5 orders	1.Place 5 orders 2.Compare Order IDs	Each order has a unique UUID; no duplicates	—
TC-M2F8-05	Negative	Checkout with empty cart	No items in cart	1.Navigate to checkout with empty cart	Checkout button disabled; message: 'Your cart is empty.'	—
TC-M2F8-06	Negative	Checkout without delivery address	Items in cart, no address	1.Tap Checkout 2.Skip address selection	Error: 'Please select a delivery address to continue.'	—
TC-M2F8-07	Negative	Double-tap Confirm Order button	Submit button tapped twice rapidly	1.Reach confirmation 2.Double-tap Confirm Order	Idempotency key prevents duplicate order; only 1 order created	—
TC-M2F8-08	Negative	Session expires during checkout	Session timeout mid-checkout	1.Begin checkout 2.Let session expire 3.Tap Confirm	Redirect to Login; cart preserved in local state; no ghost order	—
TC-	UI/UX	Order summary shows full	Final confirmation	1.Reach order confirmation step	Items, quantities, prices, delivery fee,	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
M2F8-09		breakdown before confirm	screen	2.Review	promo, total all visible	
TC-M2F8-10	Performance	Order placement completes in <5 seconds	Single checkout on 4G	1.Tap Confirm Order 2.Measure until confirmation screen	Order confirmed and screen displayed within 5 seconds	—

2.9 M2-F9: Order Cancellation

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Cancel Window	Within 2 minutes of placing order	After 2 minutes, after preparation started	Exactly at 2-minute mark	2-minute cancellation window
Reason	Changed mind, wrong address, duplicate	Null reason, empty string	N/A	Reason required for tracking
Refund	Paid via JazzCash/Card — refund initiated	COD — no refund needed	Partial refund (promo applied)	Refund within 2 business days
Active Order	Status: Confirmed (not yet Preparing)	Status: Preparing, On the Way	Status: Confirmed at exactly 2 min	Cannot cancel after Preparing

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M2F9-01	Functional	Cancel order within 2-minute window	Order placed 1 min ago, status: Confirmed	1.Open active order 2.Tap Cancel 3.Select reason 4.Confirm	Order cancelled; refund initiated; cancellation confirmation shown	—
TC-M2F9-02	Functional	Cancellation reason submitted	Reason: 'Wrong delivery address'	1.Tap Cancel 2.Select reason from list 3.Confirm	Reason recorded; visible in order history as cancellation note	—
TC-M2F9-03	Functional	Refund initiated for JazzCash paid order	Cancelled paid order	1.Cancel order paid via JazzCash	Refund status shown: 'Refund initiated; expected in 2 business days'	—
TC-M2F9-04	Functional	COD order cancellation — no	Cancelled COD order	1.Cancel COD order within window	Order cancelled; no refund needed; status:	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
		refund needed			Cancelled	
TC-M2F9-05	Negative	Cancel order after 2-minute window	Order placed 3 min ago	1.Try to cancel 3 min after placing	Cancel button disabled; message: 'Cancellation window has passed.'	—
TC-M2F9-06	Negative	Cancel order already in Preparing status	Status: Preparing	1.Open active order in Preparing state 2.Try to cancel	Error: 'Your order is already being prepared and cannot be cancelled.'	—
TC-M2F9-07	Negative	Cancel with no reason selected	Skip reason selection	1.Tap Cancel 2.Skip reason 3.Confirm	Error: 'Please select a reason for cancellation.'	—
TC-M2F9-08	UI/UX	Cancel option not visible after window expires	Order >2 minutes old	1.Open order 2+ minutes after placing	Cancel button hidden or visibly disabled with explanation	—
TC-M2F9-09	UI/UX	Cancellation countdown timer shown	Active cancellable order	1.Open active order within 2 min	Countdown timer '1:45 to cancel' visible; creates urgency	—
TC-M2F9-10	Security	User can only cancel own orders	User A cancels User B's order via API	1.Get User B's order ID 2.Send cancel request as User A	403 Forbidden; cancellation rejected	—

2.10 M2-F10: Nutritional Info & Allergen Display

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Item Type	Beef Burger, Biryani, Classic Pizza	Non-existent item ID	Item with 0 calories (beverage)	All items have nutrition data
Nutrient Data	Calories, Protein, Fat, Carbs, Sodium	Missing field (null)	0g fat, 0g protein edge cases	Per serving values
Allergens	Gluten, Dairy, Nuts, Eggs, Soy	Unknown allergen string	Item with all 5 allergens	Highlighted list per item
Language	English and Urdu nutrient labels	Unsupported language	Both languages at once	Bilingual support

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M2F10-01	Functional	View nutritional info for Beef Burger	Item: Beef Burger	1.Open Beef Burger 2.Tap Nutritional Info	Calories, protein, fat, carbs, sodium displayed per serving	—
TC-M2F10-02	Functional	Allergen list displayed for item	Item: Classic Pizza	1.Open Classic Pizza 2.View allergens	Allergens listed: Gluten, Dairy; clearly labelled with icons	—
TC-M2F10-03	Functional	Nutritional info in Urdu	Language: Urdu, open any item	1.Set Urdu 2.Open Biryani 3.View nutrition	Nutrient labels in Urdu (, ,)	—
TC-M2F10-04	Functional	Zero-calorie item shows 0 kcal	Item: Plain water or diet soda	1.Open zero-calorie item 2.Check nutrition	0 kcal displayed; not blank or null	—
TC-M2F10-05	Negative	Nutritional data missing for an item (null)	Item with no nutrition data in DB	1.Open item with missing nutrition	'Nutritional information not available for this item.' — no crash	—
TC-M2F10-06	Negative	Unknown allergen returned from API	API returns allergen: 'UNICORN'	1.View item allergens 2.Unknown allergen in response	Unknown allergen displayed as plain text; no crash; no icon	—
TC-M2F10-07	UI/UX	Allergen icons visually distinct	Item with allergens	1.Open item 2.View allergen section	Each allergen has distinct icon + label (not text-only).	—
TC-M2F10-08	UI/UX	Nutritional info accessible without adding to cart	Browse menu, not logged in	1.Open any item without logging in	Nutrition tab visible and tappable; no login required	—
TC-M2F10-09	UI/UX	Nutrition panel scrollable if many nutrients	Item with 10+ nutritional attributes	1.Open nutrition panel	Panel scrolls smoothly; no truncation; all values visible	—
TC-M2F10-10	Performance	Nutrition data loads in <2 seconds	Open item detail page	1.Tap item 2.Measure time for nutrition tab to load	Nutritional info fully loaded within 2 seconds	—

3. Module M3: Payment Processing (FE-7)

Covers all payment-related features: JazzCash wallet payments, Easypaisa mobile account payments, credit/debit card processing, Cash on Delivery, saved payment methods, Spice Points redemption, refunds, payment receipts, transaction history, and payment security.

3.1 M3-F1: JazzCash Payment

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Account Number	03001234567 (active JazzCash)	03001234567 wrong PIN, non-JazzCash no.	Account with exact order amount	Pakistan 11-digit mobile
MPIN	4-digit MPIN: 1234	Wrong MPIN, too short (3 digit), letters	MPIN with leading zero: 0123	4-digit numeric
Balance	Balance > order amount	Balance < order amount (insufficient)	Balance exactly = order amount	Real-time balance check
Order Amount	PKR 100 – PKR 9,999	PKR 0, PKR 10,001 (JazzCash limit)	PKR 10,000 (max per transaction)	Daily limit applies

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M3F1-01	Functional	Successful JazzCash payment	Account: 03001234567, MPIN: 1234, Amt: PKR 450	1.Select JazzCash 2.Enter number 3.Enter MPIN 4.Confirm	Payment successful; order confirmed; Transaction ID shown	—
TC-M3F1-02	Functional	JazzCash payment confirmation SMS received	Valid JazzCash payment	1.Complete JazzCash payment	JazzCash debit SMS received within 30 sec	—
TC-M3F1-03	Functional	JazzCash payment for order at PKR 10,000 limit	Order exactly PKR 10,000	1.Build PKR 10,000 order 2.Pay via JazzCash	Payment succeeds at limit; no error	—
TC-M3F1-04	Negative	JazzCash payment with wrong MPIN (3 attempts)	Wrong MPIN 3 times	1.Enter wrong MPIN 3 times	After 3rd: 'MPIN blocked. Please reset via JazzCash app.'	—
TC-M3F1-05	Negative	Insufficient JazzCash balance	Order PKR 1,200, Balance PKR 500	1.Attempt JazzCash payment	Error: 'Insufficient JazzCash balance. Please top up or	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
					choose another method.'	
TC-M3F1-06	Negative	Order amount exceeds JazzCash per-transaction limit	Order PKR 11,000	1.Try to pay PKR 11,000 via JazzCash	Error: 'JazzCash limit is PKR 10,000 per transaction. Please split or use card.'	—
TC-M3F1-07	Negative	Non-JazzCash number entered	Telenor number: 03441234567	1.Enter Telenor number as JazzCash	Error: 'This number is not registered with JazzCash.'	—
TC-M3F1-08	UI/UX	JazzCash logo and branding visible on payment screen	Payment method selection	1.Open payment methods	JazzCash logo, correct brand orange colour, 'Mobile Account' label shown	—
TC-M3F1-09	Security	JazzCash MPIN not stored by SpiceSync	Complete JazzCash payment	1.Pay via JazzCash 2.Inspect app DB/logs	MPIN never stored in SpiceSync DB or logs; passed directly to JazzCash API	—
TC-M3F1-10	Performance	JazzCash API responds in <5 seconds	Normal load single transaction	1.Confirm JazzCash payment 2.Measure response	Gateway responds and order confirmed within 5 seconds	—

3.2 M3-F2: Easypaisa Payment

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Account	03211234567 (active Easypaisa)	Non-Easypaisa number, inactive account	Account at zero balance	Telenor-based mobile account
OTP	6-digit OTP from SMS: 654321	Wrong OTP, expired OTP (>3 min), letters	OTP at exactly 2:59 vs 3:01	OTP expires in 3 min
Balance	Balance >= order amount	Balance < order amount	Balance exactly = order amount	Real-time
Order Amount	PKR 100 – PKR 25,000	PKR 0, PKR 25,001	PKR 25,000 (Easypaisa max)	Per-transaction limit

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M3F2-01	Functional	Successful Easypaisa payment with OTP	Account: 03211234567, OTP: 654321	1.Select Easypaisa 2.Enter number 3.Enter SMS OTP 4.Confirm	Payment successful; order confirmed	—
TC-M3F2-02	Functional	OTP resend after expiry	OTP expired, tap Resend	1.Request OTP 2.Wait 4 min 3.Tap Resend	New OTP sent; old OTP invalidated; 3-min timer resets	—
TC-M3F2-03	Functional	Easypaisa SMS confirmation received	Completed payment	1.Pay via Easypaisa 2.Check SMS	Easypaisa debit SMS received within 30 sec	—
TC-M3F2-04	Negative	Wrong Easypaisa OTP entered	Wrong OTP: 999999	1.Enter wrong OTP 2.Tap Verify	Error: 'Incorrect OTP. X attempts remaining.'	—
TC-M3F2-05	Negative	Easypaisa OTP entered after 3-minute expiry	OTP entered at 3 min 30 sec	1.Request OTP 2.Wait 4 min 3.Enter OTP	Error: 'OTP has expired. Please request a new code.'	—
TC-M3F2-06	Negative	Insufficient Easypaisa balance	Order PKR 800, Balance PKR 200	1.Enter Easypaisa number 2.Confirm payment	Error: 'Insufficient balance. Please top up your Easypaisa account.'	—
TC-M3F2-07	Negative	Amount exceeds Easypaisa transaction limit (PKR 25,000)	Order PKR 26,000	1.Try to pay PKR 26,000	Error: 'Easypaisa max transaction is PKR 25,000. Use another payment method.'	—
TC-M3F2-08	UI/UX	OTP input field auto-focuses after number entry	Easypaisa payment flow	1.Enter account number 2.Tap Continue	OTP input field auto-focused; numeric keyboard shown	—
TC-M3F2-09	Security	Easypaisa OTP not stored in app logs	Complete Easypaisa payment	1.Pay via Easypaisa 2.Check app logs	OTP value not present in any app logs or analytics events	—
TC-M3F2-10	Performance	50 concurrent Easypaisa transactions	50 simultaneous payments	1.Simulate 50 concurrent Easypaisa payments	All transactions complete; no duplicate charges; response <8s	—

3.3 M3-F3: Credit/Debit Card Payment

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
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Category	Valid Data	Invalid Data	Boundary	Notes
Card Number	Visa: 4111111111111111, MC: 55000000000000004	Luhn-fail: 4111111111111112, too short	16-digit card number boundary	Luhn algorithm validated
Expiry Date	12/27, 06/26	01/20 (past), 13/26 (invalid month)	Month=12, Year=current year	MM/YY format
CVV	Visa CVV: 123, Amex CVV: 1234	00, 1234 for Visa, letters, empty	3 digits (Visa/MC), 4 (Amex)	Never stored
3DS	3DS challenge completed	3DS timeout, user cancels 3DS	3DS redirect and return	Handled by bank

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M3F3-01	Functional	Successful Visa card payment	Visa: 4111111111111111, Exp:12/27, CVV:123	1.Select Card 2.Enter details 3.Complete 3DS if prompted 4.Confirm	Payment authorised; order placed; confirmation shown	—
TC-M3F3-02	Functional	Save card for future orders	Valid Visa card, 'Save this card' checked	1.Pay with card 2.Check Save card option 3.Confirm	Card tokenised and saved; CVV not stored; shown in next checkout	—
TC-M3F3-03	Functional	Pay with previously saved card	Saved Visa card	1.Open checkout 2.Select saved card 3.Enter CVV 4.Confirm	Payment processes; no need to re-enter card number	—
TC-M3F3-04	Functional	3DS challenge completed successfully	Bank requires 3DS verification	1.Enter card 2.Complete 3DS challenge in bank popup 3.Return	3DS passed; payment authorised; order confirmed	—
TC-M3F3-05	Negative	Expired credit card	Exp: 01/2020	1.Enter expired card 2.Tap Pay	Error: 'Your card has expired. Please use a valid card.'	—
TC-M3F3-06	Negative	Invalid card number (Luhn check fails)	Card: 4111111111111112	1.Enter Luhn-failing card number	Error: 'Invalid card number.' Submit blocked client-side	—
TC-M3F3-07	Negative	Wrong CVV	Correct card, CVV: 000	1.Enter valid card with wrong CVV 2.Submit	Bank declines; 'Card declined. Please check your details or use another card.'	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M3F3-08	Negative	3DS challenge cancelled by user	User closes 3DS popup	1.Start card payment 2.Close 3DS window	Payment cancelled; user returns to payment screen; order not placed	—
TC-M3F3-09	UI/UX	Card type auto-detected from first 4 digits	Enter 4111... (Visa) or 5500... (MC)	1.Start typing card number	Visa logo or Mastercard logo appears automatically as first digits entered	—
TC-M3F3-10	Security	CVV never stored in SpiceSync database	Complete card payment	1.Pay with card 2.Inspect DB records	CVV field absent from all DB tables; PCI-DSS compliant storage confirmed	—

3.4 M3-F4: Cash on Delivery (COD)

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Supported Area	Islamabad, Lahore, Karachi urban zones	Remote/rural areas, non-supported cities	Edge of delivery zone	COD not available everywhere
Order Limit	PKR 100 – PKR 5,000	PKR 5,001+ (exceeds COD cap)	Exactly PKR 5,000 (at cap)	PKR 5,000 COD maximum
Exact Change	Customer informed to have exact change	N/A	N/A	Rider may not carry change
Delivery Agent	Assigned rider with COD capability	Rider not COD-enabled	N/A	Rider assignment logic

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M3F4-01	Functional	COD order placed in Islamabad E-11	City: Islamabad, Order: PKR 800, COD	1.Select Islamabad E-11 2.Choose COD 3.Confirm	Order placed; 'Pay on Delivery' badge; rider assigned	—
TC-M3F4-02	Functional	COD order shows exact-change notice	Any COD order	1.Select COD 2.Review summary	Notice displayed: 'Please have exact change of PKR X ready.'	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M3F4-03	Functional	COD order at PKR 5,000 limit	Order exactly PKR 5,000	1.Build PKR 5,000 cart 2.Select COD	COD option available and functional at exact limit	—
TC-M3F4-04	Negative	COD for order exceeding PKR 5,000	Order total: PKR 6,500	1.Build PKR 6,500 cart 2.Try COD	COD option disabled; tooltip: 'COD available for orders up to PKR 5,000 only.'	—
TC-M3F4-05	Negative	COD selected in unsupported area	City: Abbottabad outskirts (rural)	1.Select unsupported area 2.Choose COD	COD option greyed out with 'Not available in this area' label	—
TC-M3F4-06	Negative	Place COD order without delivery address	No address, COD selected	1.Select COD 2.Tap Order Now without address	Error: 'Please add a delivery address.'	—
TC-M3F4-07	UI/UX	COD option shows 'Cash on Delivery' label with cash icon	Payment screen	1.Open payment methods	COD clearly labelled with cash/notes icon; not just 'Other'	—
TC-M3F4-08	UI/UX	COD unavailable areas show tooltip explaining why	Payment screen, unsupported area	1.Select unsupported area 2.View payment options	COD shown in list but greyed; tap shows tooltip reason	—
TC-M3F4-09	Security	COD order amount not modifiable post-placement	COD order placed	1.Place COD order 2.Intercept status update 3.Try to change amount	Amount locked server-side; 403 if modification attempted	—
TC-M3F4-10	Performance	COD order confirmation in <5 seconds	COD checkout flow	1.Confirm COD order 2.Measure time	Order confirmed and screen updated in <5 seconds	—

3.5 M3-F5: Saved Payment Methods

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Saved Cards	Up to 3 saved cards	4th card (exceeds limit)	Exactly 3 cards	Max 3 saved cards
Tokenisation	Card token (no PAN stored)	Invalid/expired token	Token validity = card expiry	PCI-DSS tokenisation
Delete Method	Delete saved card	Delete non-existent card	Last saved card	Confirmation required

Category	Valid Data	Invalid Data	Boundary	Notes
		ID	deletion	
Default Method	Set 1 card as default	Setting 2 cards as default	N/A	Single default allowed

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M3F5-01	Functional	Save credit card after first use	Visa card, Save checked	1.Pay with new Visa 2.Check Save Card 3.Complete payment	Card token saved; appears in Saved Methods list	—
TC-M3F5-02	Functional	Use saved card for next order	Saved Visa in list	1.Open checkout 2.Select saved Visa 3.Enter CVV 4.Pay	Payment processes using saved token; success	—
TC-M3F5-03	Functional	Set saved card as default	Saved Visa card	1.Open Saved Methods 2.Set Visa as default	Visa pre-selected on next checkout; marked with 'Default' badge	—
TC-M3F5-04	Functional	Delete saved card	Saved Mastercard in list	1.Open Saved Methods 2.Tap Delete on Mastercard 3.Confirm	Card removed from list; not shown in checkout	—
TC-M3F5-05	Negative	Add 4th saved card (exceeds limit)	Already 3 cards saved	1.Try to save a 4th card	Error: 'Maximum 3 saved cards allowed. Delete one to add new.'	—
TC-M3F5-06	Negative	Pay with saved card that has expired	Saved Visa with Exp: 01/2024	1.Select expired saved card 2.Tap Pay	Error: 'This saved card has expired. Please add a new card.'	—
TC-M3F5-07	Negative	Delete non-existent saved card (API tamper)	Send delete request for card ID not owned	1.Send DELETE /cards/{fake_id}	403 Forbidden; card not deleted	—
TC-M3F5-08	UI/UX	Saved cards list shows masked card number	Saved Methods screen	1.Open Payment > Saved Methods	Cards shown as **** **** * 1111 (last 4 digits only) + expiry + brand logo	—
TC-M3F5-09	Security	Full card number never stored in SpiceSync	Save any card	1.Save card 2.Query SpiceSync DB/API for card	Only token, last 4 digits, expiry, and card type stored — no full PAN	—
TC-	Security	Saved cards	User A views User	1.Log in as User A	404 Not Found or 403	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
M3F5-10		inaccessible to other users	B's saved cards	2.Request User B's saved cards endpoint	Forbidden; no card data leaked	

3.6 M3-F6: Spice Points (Loyalty) Redemption

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Points Balance	500 points = PKR 50, 1000 = PKR 100	0 points, negative balance	Exactly 500 points (min to redeem)	100 points = PKR 10
Redemption Cap	Max PKR 200 per order	Attempt to redeem PKR 201	Exactly PKR 200 redemption	Cap per order enforced
Order Min	Min order PKR 300 to use points	Order PKR 200 (below min)	Exactly PKR 300	Order minimum for redemption
Earning	Earn 10 pts per PKR 100 spent	Points on cancelled order	First order (0 existing points)	Points expire in 6 months

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M3F6-01	Functional	Redeem 500 Spice Points for PKR 50 discount	User has 500 points, order PKR 300	1.At checkout toggle Redeem Points 2.Confirm	PKR 50 deducted; points balance reduced by 500	—
TC-M3F6-02	Functional	Earn points after successful order	Complete PKR 500 order	1.Place and receive order 2.Check points balance	50 points added (10 per PKR 100); reflected in Spice Points screen	—
TC-M3F6-03	Functional	Points redeemed + promo code stacked	EID2025 promo + 500 points	1.Apply EID2025 2.Redem points 3.Checkout	Both discounts applied; total correctly reduced	—
TC-M3F6-04	Functional	Points balance shown at checkout	Any logged-in user with points	1.Proceed to checkout	'You have X points = PKR Y' shown as redemption option	—
TC-M3F6-05	Negative	Redeem points with zero balance	User has 0 points	1.Toggle Redeem Points 2.Confirm	Error: 'You don't have enough Spice Points to redeem.'	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M3F6-06	Negative	Redeem points on order below minimum (PKR 300)	Order PKR 200, user has 500 pts	1.Add PKR 200 item 2.Try to redeem points	Error: 'Minimum order of PKR 300 required to redeem Spice Points.'	—
TC-M3F6-07	Negative	Attempt to redeem >PKR 200 cap per order	User has 5000 pts, order PKR 1000	1.Try to redeem all 5000 pts	Only PKR 200 max redeemable; system caps at 2000 points per order	—
TC-M3F6-08	Negative	Points not earned on cancelled order	Place then cancel order	1.Place order 2.Earn points 3.Cancel	Points reversed; balance returns to pre-order level	—
TC-M3F6-09	UI/UX	Spice Points balance prominently shown in profile	Profile screen, any user	1.Open Profile	Spice Points balance shown with flame icon; PKR equivalent shown	—
TC-M3F6-10	Security	Points balance tampered via API	Intercept checkout, inflate points	1.Intercept checkout request 2.Set points_to_redeem=99999	Server validates against actual balance; rejects overage	—

3.7 M3-F7: Refund Processing

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Payment Method	JazzCash paid order, Easypaisa, Card	COD order (no electronic refund)	Partial refund (promo applied)	Refund to original payment method
Cancellation	Cancelled within 2-min window	Order already delivered	Cancellation at exactly 2 min	Eligible for refund
Refund Amount	Full refund for full cancellation	Partial refund (item issue)	Refund = order total – delivery fee	Correct amount validation
Timeline	Initiated within 1 hour of cancellation	Delayed >3 business days	Exactly at 2 business days	Business days (Mon-Fri)

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
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TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M3F7-01	Functional	Refund initiated after JazzCash order cancellation	JazzCash paid order cancelled in time	1.Place order (JazzCash) 2.Cancel within window	Refund initiated; status: 'Refund Initiated'; notification sent	—
TC-M3F7-02	Functional	Refund appears in JazzCash wallet within 2 business days	Cancelled JazzCash order	1.Cancel 2.Wait 2 business days 3.Check JazzCash balance	Full amount returned to JazzCash wallet	—
TC-M3F7-03	Functional	Card payment refund processed via original card	Visa card paid order cancelled	1.Cancel card-paid order 2.Check card statement in 2 days	Refund reflected on original Visa card	—
TC-M3F7-04	Functional	Refund status visible in order history	Cancelled paid order	1.Open Order History 2.Find cancelled order	Status shows 'Refund Initiated' > 'Refund Processed'	—
TC-M3F7-05	Negative	Refund attempt for COD order	Cancel a COD order	1.Cancel COD order	No refund initiated; status: 'Cancelled (No charge)'; correct handling	—
TC-M3F7-06	Negative	Refund request for already-delivered order	Order with status Delivered	1.Attempt to get refund on delivered order	Refund not available; contact support option shown; no auto-refund	—
TC-M3F7-07	Negative	Duplicate refund request	Submit refund request twice	1.Cancel order 2.Submit refund twice	Idempotency: second request ignored; no double-refund	—
TC-M3F7-08	UI/UX	Refund status updates in order history	Refund in progress	1.Open order with 'Refund Initiated' status	Status chip visible with timestamp; progress shown	—
TC-M3F7-09	Security	Refund amount cannot be modified by user	Intercept refund API call	1.Cancel order 2.Intercept refund request 3.Inflate amount	Server calculates refund independently; client amount ignored	—
TC-M3F7-10	Performance	Refund initiation completes in <3 seconds	Cancel paid order	1.Tap Cancel 2.Confirm 3.Measure refund initiation time	Refund initiated and confirmation shown within 3 seconds	—

3.8 M3-F8: Payment Receipt Generation

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Receipt Channels	Email receipt, In-app receipt, SMS summary	No email (notification disabled)	Receipt for PKR 0 (fully discounted)	Email within 2 min, in-app instant
Receipt Contents	Order ID, items, quantities, prices, payment method, date	Missing item in receipt	Total = sum of items + delivery - discount	All fields required
Format	In-app: HTML view; Email: PDF/HTML	Corrupted email	N/A	PDF option for download
Language	English and Urdu receipts	Unsupported locale	Bilingual receipt	Language matches app setting

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M3F8-01	Functional	Receipt email received after successful payment	Any paid order	1.Complete payment 2.Check registered email	Receipt email arrives within 2 min with full order breakdown	—
TC-M3F8-02	Functional	In-app receipt accessible from Order History	Any completed order	1.Open Order History 2.Tap order 3.View Receipt	Full receipt shown: items, prices, discounts, payment method, date	—
TC-M3F8-03	Functional	Receipt download as PDF	Completed order receipt	1.Open receipt 2.Tap Download PDF	PDF receipt downloaded; opens in PDF viewer	—
TC-M3F8-04	Functional	Receipt in Urdu when language is Urdu	Language: Urdu, completed order	1.Set Urdu 2.Complete order 3.View receipt	Receipt labels, item names, and amounts displayed in Urdu	—
TC-M3F8-05	Negative	Receipt for cancelled order	Cancelled before payment	1.Cancel unpaid order 2.Check for receipt	No receipt generated; order shows as 'Cancelled — No Charge'	—
TC-M3F8-06	Negative	Receipt email not sent if email notifications disabled	Email notifications OFF	1.Disable email notifications 2.Place order	Email receipt not sent; in-app receipt still accessible	—
TC-M3F8-07	Negative	Receipt for fully discounted order (PKR 0)	100% promo code applied	1.Apply 100% discount code 2.Place order	Receipt generated showing PKR 0 total; all discounts itemised	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M3F8-08	UI/UX	Receipt clearly shows all discount lines	Order with promo + loyalty points	1.View receipt for discounted order	Separate discount lines: Promo (-PKR X), Points (-PKR Y), Total clearly shown	—
TC-M3F8-09	Security	Receipt does not expose full card number	Card-paid order receipt	1.View receipt for card payment	Receipt shows only last 4 digits of card (e.g., Visa ****1111)	—
TC-M3F8-10	Performance	In-app receipt loads in <2 seconds	Open any completed order receipt	1.Tap receipt button 2.Measure load time	Receipt fully rendered within 2 seconds	—

3.9 M3-F9: Transaction History

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
Date Range	Last 7 days, 30 days, custom range	Start date > end date, future date	Single-day range	Max 12 months history
Filter	Filter by: All, Paid, Refunded, Cancelled	Unknown status filter	All transactions in one day	Status-based filter
Record Count	1 to 100 transactions	0 transactions (empty history)	Exactly 100 (page size)	Paginated: 20 per page
Export	CSV export of history	Export with 0 records	Export all 12 months	CSV only

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M3F9-01	Functional	View last 30 days transaction history	30+ transactions in last month	1.Open Transaction History 2.Select last 30 days	All transactions listed; correct amounts, dates, payment methods	—
TC-M3F9-02	Functional	Filter by 'Refunded' transactions	2+ refunded transactions in history	1.Apply filter: Refunded	Only refunded transactions shown; count matches	—
TC-M3F9-03	Functional	Export transaction history as CSV	30-day history	1.Tap Export 2.Select CSV 3.Download	CSV file downloaded with all transactions;	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
					columns: Date, Order ID, Amount, Method, Status	
TC-M3F9-04	Functional	Paginate through transaction history	100+ transactions	1.Open history 2.Scroll to end of page 1	Next 20 transactions load smoothly on scroll (infinite scroll or pagination)	—
TC-M3F9-05	Negative	Select date range with start > end	Start: 2025-05-10, End: 2025-04-01	1.Set invalid date range 2.Tap Apply	Error: 'Start date must be before end date.'	—
TC-M3F9-06	Negative	View transaction history with no past orders	New account with no orders	1.Open Transaction History 2.Select any date range	Empty state: 'No transactions found for this period.' — no crash	—
TC-M3F9-07	Negative	Export CSV with 0 records	Date range with no transactions	1.Select date range with no orders 2.Export	CSV downloaded with header row only; no crash	—
TC-M3F9-08	UI/UX	Each transaction row shows key info	Transaction history list	1.Open history 2.View transaction rows	Each row: date, order ID (truncated), amount in PKR, status badge visible	—
TC-M3F9-09	Security	User can only view own transaction history	User A accesses User B's history URL	1.Copy User B's transaction history URL 2.Access as User A	403 Forbidden; no transaction data shown	—
TC-M3F9-10	Performance	History loads 20 transactions in <2 seconds	Default view on load	1.Open Transaction History 2.Measure initial load	First 20 transactions displayed within 2 seconds	—

3.10 M3-F10: Payment Security & Fraud Prevention

Test Data

Category	Valid Data	Invalid Data	Boundary	Notes
TLS Version	TLS 1.2 or 1.3	TLS 1.0, SSLv3	Exactly TLS 1.2 (minimum)	All payment APIs enforce TLS
HTTPS	All payment endpoints HTTPS	HTTP payment request	N/A	HTTP requests blocked
Fraud Patterns	Single transaction, normal amount	Rapid 10 payments in 1 min, abnormal location	Exactly 10 transactions/min	ML-based fraud detection

Category	Valid Data	Invalid Data	Boundary	Notes
			(threshold)	
Data in Logs	No PAN, CVV, MPIN in logs	Partial card number in error log	N/A	PCI-DSS requirement

Test Cases

TC-ID	Type	Description	Test Data	Steps	Expected	Status
TC-M3F10-01	Functional	All payment API calls use HTTPS	Any payment transaction	1.Intercept payment API traffic via proxy	All calls use HTTPS; 0 HTTP payment requests observed	—
TC-M3F10-02	Functional	TLS 1.2+ enforced for payment endpoints	Network inspection	1.Test payment endpoints with TLS 1.0 client	TLS 1.0 connections rejected; minimum TLS 1.2 accepted	—
TC-M3F10-03	Functional	Certificate pinning prevents MITM	MITM proxy with self-signed cert	1.Intercept with self-signed cert via Burp Suite	App detects certificate mismatch; payment blocked; error shown	—
TC-M3F10-04	Functional	Fraud alert triggered on suspicious velocity	10 transactions in 1 minute from same account	1.Rapid-fire 10 payments in 60 seconds	Fraud flag raised; account temporarily restricted; admin alerted	—
TC-M3F10-05	Negative	Payment over HTTP (non-HTTPS) blocked	Force HTTP payment request	1.Send payment request over HTTP	Request blocked; 301 redirect to HTTPS or 403; payment not processed	—
TC-M3F10-06	Negative	SQL injection in payment amount field	Amount: '100; DROP TABLE payments;'	1.Enter SQL injection as amount 2.Submit	Input sanitised; payment rejected; no DB error	—
TC-M3F10-07	Negative	Replay attack — reuse captured payment token	Capture valid payment request, replay it	1.Capture JazzCash payment request 2.Replay	Idempotency key rejected; 'Transaction already processed'	—
TC-M3F10-08	Negative	Abnormal amount (negative or zero) blocked server-side	Amount: -500, Amount: 0	1.Send negative amount via API 2.Send zero amount	Both rejected with 400 Bad Request; no transaction created	—
TC-M3F10-09	Security	No PAN or CVV in application logs	Complete card payment, inspect logs	1.Pay with card 2.Inspect app/server logs	No card number or CVV in any log entry; PCI-DSS compliant	—
TC-	Performance	Fraud check does	Normal single	1.Complete payment	Fraud check is non-	—

TC-ID	Type	Description	Test Data	Steps	Expected	Status
M3F10-10		not add >500ms to payment flow	transaction	2.Measure total time including fraud check	blocking; adds <500ms to payment flow	

4. Test Summary Matrix

Total: 3 Modules × 10 Features × 10 Test Cases = 300 Test Cases

Module / Feature	Functional	Negative	UI/UX	Security	Performance	Total
Module 1: User Account Management	44	30	14	7	5	100
Module 2: Order Placement & Cart	46	28	14	5	7	100
Module 3: Payment Processing (FE-7)	38	32	13	11	6	100
GRAND TOTAL	128	90	41	23	18	300